

Circulatory limits to oxygen supply during an acute temperature increase in the Chinook salmon (*Oncorhynchus tshawytscha*)

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¹Faculty of Land and Food Systems, University of British Columbia, Vancouver, Canada; ²Department of Zoology, University of Gothenburg, Gothenburg, Sweden; ³Department of Zoology, University of British Columbia, Vancouver, Canada; and ⁴Department of Forest Sciences, University of British Columbia, Vancouver, Canada

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Clark TD, Sandblom E, Cox GK, Hinch SG, Farrell AP. Circulatory limits to oxygen supply during an acute temperature increase in the Chinook salmon (*Oncorhynchus tshawytscha*). *Am J Physiol Regul Integr Comp Physiol* 295: R1631–R1639, 2008. First published September 3, 2008; doi:10.1152/ajpregu.90461.2008.— This study was undertaken to provide a comprehensive set of data relevant to disclosing the physiological effects and possible oxygen transport limitations in the Chinook salmon (*Oncorhynchus tshawytscha*) during an acute temperature change. Fish were instrumented with a blood flow probe around the ventral aorta and catheters in the dorsal aorta and sinus venosus. Water temperature was progressively increased from 13°C in steps of 4°C up to 25°C. Cardiac output increased from 29 to 56 ml·min⁻¹·kg⁻¹ between 13 and 25°C through an increase in heart rate (58 to 105 beats/min). Systemic vascular resistance was reduced, causing a stable dorsal aortic blood pressure, yet central venous blood pressure increased significantly at 25°C. Oxygen consumption rate increased from 3.4 to 8.7 mg·min⁻¹·kg⁻¹ during the temperature increase, although there were signs of anaerobic respiration at 25°C in the form of increased blood lactate and decreased pH. Arterial oxygen partial pressure was maintained during the heat stress, although venous oxygen partial pressure (PvO₂) and venous oxygen content were significantly reduced. Cardiac arrhythmias were prominent in three of the largest fish (>4 kg) at 25°C. Given the switch to anaerobic metabolism and the observation of cardiac arrhythmias at 25°C, we propose that the cascade of venous oxygen depletion results in a threshold value for PvO₂ of around 1 kPa. At this point, the oxygen supply to systemic and cardiac tissues is compromised, such that the oxygen-deprived and acidotic myocardium becomes arrhythmic, and blood perfusion through the gills and to the tissues becomes compromised.

blood pressure; blood respiratory properties; cardiac output; hematocrit; hemoglobin; heart rate; blood plasma; rate of oxygen consumption; oxygen equilibrium curves; Pacific salmon; stroke volume; vascular resistance

WITH CURRENT TRENDS IN CLIMATE change, it is becoming increasingly important to understand both the behavioral and physiological responses of animals to perturbations in temperature (41). The highly efficient counter-current arrangement between blood and respiratory water at fish gills means that any acute change in ambient water temperature rapidly changes body temperature and modifies tissue oxygen requirements (10–13, 43, 58). A change in the rate of oxygen consumption ($\dot{M}_{O_2}$) then requires a parallel change in oxygen delivery from the gills to tissues by the circulatory system, which is determined by the product of cardiac output (Q) and the difference between

arterial (CaO₂) and venous (CvO₂) oxygen content (CaO₂–CvO₂) according to the Fick equation:

$$\dot{M}_{O_2} = Q \cdot (CaO_2 - CvO_2) \quad (1)$$

Q is the product of heart rate (f_H) and cardiac stroke volume (V_S), while CaO₂–CvO₂ is dependent on oxygen partial pressure and capacitance of the blood for oxygen. Blood oxygen capacitance itself is reliant on mean corpuscular hemoglobin concentration (MCHC), oxygen binding characteristics, and the degree of saturation of the hemoglobin molecule (62). Thus, changes in circulatory oxygen delivery resulting from variations in body temperature can be modulated by changes in one or any combination of these factors (18). In the present study, we were particularly interested in circulatory oxygen delivery because it has been suggested that circulatory performance of fishes may be the first process to cause oxygen deficiency during heat stress (40, 50). However, while many studies have investigated components of the cardiorespiratory response to heat stress in fish, the composite response has rarely been measured in a single study. Furthermore, a range of responses are reported, perhaps reflecting species variability and/or subtle but important differences in the experimental protocols.

The primary cardiovascular responses to an acute increase in water temperature involve an increase in Q , which is typically mediated by a rise in f_H . Stroke volume has been reported to either remain unchanged (8, 12, 29, 57) or decrease (3, 7, 48). CaO₂–CvO₂ is typically reported to remain unchanged or is increased due to a reduction in CvO₂ because of greater tissue oxygen extraction (33, 50). An increase in CaO₂ may also contribute to an increase in CaO₂–CvO₂, resulting from greater gill ventilation and an associated increase in arterial oxygen partial pressure (PaO₂), and/or an increase in circulating erythrocytes following splenic contraction (48). Substantial decreases in venous oxygen partial pressure (PvO₂) and CvO₂ are of concern for proper cardiac function, given that the heart (specifically, the spongy myocardium) is the last organ to be supplied with oxygen from systemic blood (14, 17).

While interspecific variability may be expected in the circulatory response to an acute temperature increase, the differences documented for the same species are surprising. Recent studies with Atlantic cod (*Gadus morhua*) provide a good example of divergent findings. Lannig et al. (36) found that dorsal aortic blood flow in North Sea Atlantic cod, as measured indirectly by flow-weighted magnetic resonance imaging, did not match the continued increase in f_H with temperature but

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rather yielded a hyperbolic regression, reflecting a strong increase in blood flow in the lower and a moderate increase in the upper temperature range. Conversely, Gollock et al. (29) found that Q , measured directly with Transonic flow probes on the ventral aorta, increased in parallel with f_H in Newfoundland Atlantic cod until water temperature reached the animals' critical thermal maximum. On the basis of oxygen equilibrium curves, they suggested, as did Heath and Hughes (33) for rainbow trout, that a decreased blood oxygen binding capacity hampered metabolic rate at high temperatures (29) and that cardiac function, therefore, was unlikely to limit aerobic metabolism in cod. Lacking from Gollock et al.'s study (29), however, were in vivo measurements of blood oxygen content to substantiate this claim. Indeed, the difficulty associated with obtaining simultaneous arterial and venous blood samples from cannulated vessels is one of the primary reasons why very few studies have concurrently investigated the full suite of cardiorespiratory variables and in vivo blood oxygen binding properties during an acute temperature change in fish. Studies of thermal limitations must necessarily measure a range of variables pertinent to circulatory function to discover the relationships between inadequate tissue oxygen delivery and circulatory collapse proposed for heat stress.

Consequently, the present study was undertaken to provide a comprehensive set of data relevant to disclosing the physiological effects and possible oxygen transport limitations in a fish during an acute temperature change. Here, we simultaneously examined the cardiovascular, hematological and respiratory responses of adult Chinook salmon (*Oncorhynchus tshawytscha*) belonging to the Chilliwack River population in British Columbia, Canada. The large size of the individuals used here permitted repetitive blood sampling as temperature was increased incrementally. This species has a coronary circulation that supplies oxygen to a portion of the ventricle (the compact myocardium), while the venous blood being pumped by the heart supplies oxygen to the atrium and the remainder of the ventricle (the spongy myocardium). The results likely have direct ecological relevance for this population of Chinook salmon since they expose themselves to rapid and broad changes in water temperature during their migration from the open ocean to freshwater spawning grounds during summer months (5, 6, 18). While juvenile Chinook salmon have been described as the most tolerant of the Pacific salmon species to elevated temperatures (6), comparatively little is known of the high-temperature tolerance of adults, except that adult fish tend to have upper lethal temperatures a few degrees lower than their juvenile counterparts (38).

MATERIALS AND METHODS

Animals and holding conditions. Ten male Chinook salmon (*Oncorhynchus tshawytscha*) in the size range of 2.1–5.4 kg body mass and 58–79-cm fork length were caught by dip-net in August 2007 as they completed their migration from the ocean to their spawning site at the Chilliwack River Hatchery, British Columbia, Canada (~100 km upstream). They were transported 20 km by truck to the Cultus Lake Salmon Research Laboratory and held outdoors in 10,000-liter circular tanks containing circulating lake water at a temperature of 13–14°C. Temperature control was achieved by mixing lake water from two different depths [7 m (21°C) and 40 m (9°C)]. The holding temperature was chosen to match the temperature conditions in the Chilliwack River at the time of capture. The fish were

not fed while in captivity (2 wk; migrating Chinook salmon do not normally feed *en route* to spawning grounds). All protocols were approved by the Animal Care Committee of the University of British Columbia in accordance with the Canadian Council on Animal Care (A05-0007).

Surgical procedures. Individual fish were netted from the holding tanks and anesthetized in water containing tricaine methanesulfonate (MS-222; 100 mg/l; Sigma, St. Louis, MO) buffered with sodium bicarbonate (NaHCO_3 ; 200 mg/l). They were weighed and placed on wet foam on a surgery table. The gills were continuously irrigated with cooled (9–11°C) and aerated water containing NaHCO_3 -buffered MS-222 at a lower dose (150 mg/l and 75 mg/l, respectively). The ventral aorta was dissected free in the isthmus anterior to the pericardium using sharp and blunt dissection and an appropriately sized blood flow probe (Transonic Systems, Ithaca, NY) was fitted around the vessel (12, 46). Great care was taken to avoid damaging the pericardium during this procedure. The ductus of Cuvier was cannulated with a polyethylene (PE)-50 catheter advanced toward the sinus venosus using the method described by Sandblom et al. (49), and the dorsal aorta was cannulated with PE-50 catheter using the method described by Soivio et al. (54). The catheters were filled with heparinized (150 IU/ml) saline and were sutured together with the flow probe lead to the skin of the fish with several silk sutures.

Following surgery, each fish was transferred to either 1) an opaque experimental holding tube (length, 120 cm; diameter, 30 cm; wire mesh at each end, with a slit in the top to externalize catheters and the flow probe lead) submersed in a larger outer tank ($n = 6$), or 2) one of two Brett-type respirometers [as described in (22, 37)] ($n = 4$). The holding tubes and respirometers were covered to minimize visual disturbance of the fish. Water temperature in the holding tubes was controlled by balancing the inflow of shallow and deep lake water, as described above, and a heating element was used to assist in obtaining the highest experimental temperature of 25°C. The respirometers were equipped with their own 9-kW titanium heating elements [(M. F. Steinhausen, E. Sandblom, E. J. Eliason, C. Verhille, A. P. Farrell, unpublished data)]. The temperature for post-surgical recovery was the same as the holding temperature (13°C). Water velocity both through the holding tubes and the swim respirometers was maintained at 20–30 cm/s for the entire experimental period, the former by utilizing submersible pumps and the latter by utilizing an inbuilt motor and propeller. The fish were allowed to recover from surgery overnight (>10 h).

Experimental protocol. Experiments began in the morning with routine cardiovascular variables being recorded for 1 h at 13°C. Oxygen consumption rate ($\dot{M}_{O_2}$) was measured during the final 20-min period of the stable temperature exposure, and blood samples were taken from the arterial and venous catheters at the end of this period. Afterward, a new temperature was attained by gradually raising the temperature 4°C over 1 h. Water temperature was stabilized for 1 h at each of 17, 21, and 25°C, during which time the cardiorespiratory measurements were repeated. Following the final temperature challenge at 25°C, the temperature was decreased over 1 h to a stable temperature of 13°C. Recordings of the recovering fish were repeated after 1 h and 2 h at 13°C following the heat stress. Upon conclusion of experiments, fish were euthanized by a cranial blow. Hearts were excised, emptied of blood, dried, weighed, and placed in 70% ethanol for subsequent separation and quantification of the two (compact and spongy) ventricular myocardial layers (for details, see Ref. 23).

Measured and calculated variables. Blood flow and blood pressure measurements were made on all 10 fish. The blood flow probe was connected to a Transonic blood flow meter (Model T206; Transonic Systems) to continuously record cardiac output (Q), and flows were corrected for temperature following experiments by using a thermostatically controlled water bath and the methods outlined in the product user manual (temperature-correction increased flows by 8–20%). Central venous (P_{CV}) and dorsal aortic (P_{DA}) blood catheters were connected to pressure transducers (model DPT-6100, pnb;

Medizintechnik, Kirchseon, Germany) that were calibrated against static water columns. The signals from the transducers were amplified using a 4-channel amplifier (Somedic AB, Hörby, Sweden). Blood flow and blood pressure signals were sampled at 50 Hz using a Biopac module (Biopac Systems, Santa Barbara, CA) connected to a laptop computer. Heart rate (f_H) was calculated from blood flow traces from the flow probe. Cardiac stroke volume (V_S) was calculated as Q/f_H . Systemic vascular resistance (R_{SYS}) was calculated from the drop in blood pressure between the dorsal aorta and the central veins as $(P_{DA}-P_{CV})/Q$.

The rate of oxygen removal from the water (=oxygen consumption rate; $\dot{M}_{O_2}$) over a 10–20 min period was obtained for the four fish that were placed in the water-tight respirometer swim tunnels. $\dot{M}_{O_2}$ was calculated from the decline in water oxygen saturation during sealed periods (accounting for temperature effects on oxygen capacitance), and the respirometers were flushed with fresh, filtered (sand and ultraviolet filtration), and aerated water at all times in between $\dot{M}_{O_2}$ measurements. The respirometers were bleached prior to the introduction of fish to preclude significant microbial oxygen consumption.

Blood samples (0.3–0.8 ml from each catheter) were collected anaerobically into heparinized syringes and immediately refrigerated for subsequent blood analysis within 1 h. The cumulative blood volume collected throughout the experimental period never

exceeded 10% of the assumed total blood volume of 4% of fish body mass (39). Occasionally, a blockage in the catheter prevented blood sampling, although at least seven fish were represented at all temperatures. Blood pH and oxygen partial pressure (P_{O_2}) were measured using a blood gas monitor (PHM 73, Radiometer, Copenhagen, Denmark) calibrated at each experimental temperature, and maintained at each temperature, using thermostatically controlled water jackets around the electrodes. Blood oxygen content was determined using a Tucker chamber (61) maintained at 37°C. Hematocrit (Hct) was determined using microhematocrit capillary tubes spun at 10,000 g for 7 min. Hemoglobin concentration ([Hb]) was always obtained using a handheld hemoglobin analyzer appropriately calibrated for fish blood [Hemocue 201⁺, www.hemocue.com; (10)], and it was sometimes also measured spectrophotometrically at 540 nm using a solution of 10 μ l of blood in 1 ml of Drabkin's solution (10, 15). The mean corpuscular hemoglobin concentration (MCHC) was calculated as $[Hb]/(Hct/100)$. Remaining whole blood was subsequently spun in Eppendorf tubes at 7,000 g for 7 min, and then the plasma was collected, frozen in liquid nitrogen, and placed at -80°C for subsequent analyses. Plasma measurements were made of cortisol (Neogen ELISA with Molecular Devices Spectramax 240pc plate reader), lactate (YSI 2300 stat plus analyzer), osmolality (Advanced

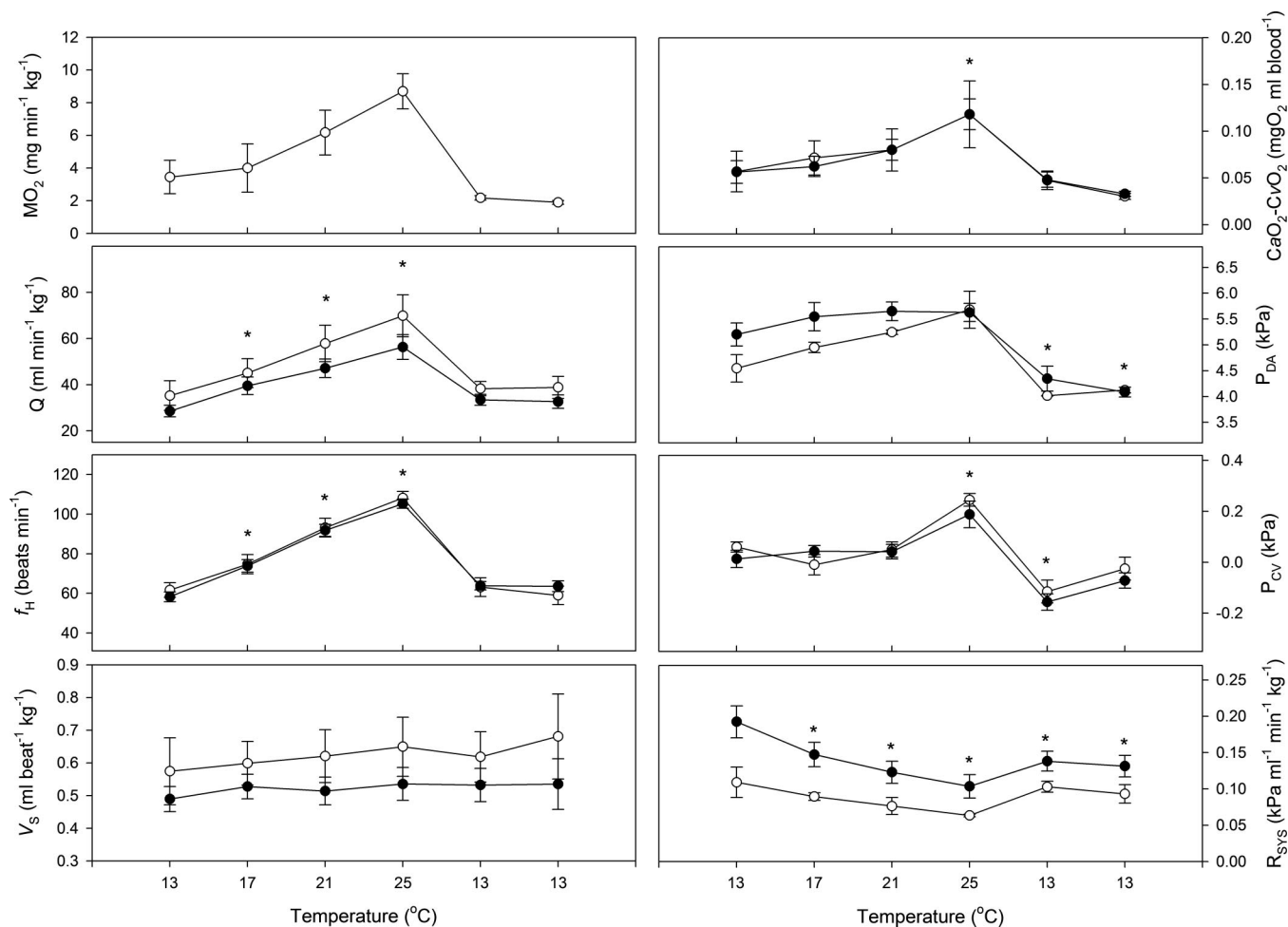


Fig. 1. Effects of acute temperature changes on cardiorespiratory variables of Chinook salmon (*Oncorhynchus tshawytscha*), including oxygen consumption rate ($\dot{M}_{O_2}$), cardiac output (Q), heart rate (f_H), stroke volume (V_S), arteriovenous oxygen content difference (CaO_2-CvO_2), dorsal aortic blood pressure (P_{DA}), central venous blood pressure (P_{CV}), and systemic vascular resistance (R_{SYS}). Dark symbols and lines include data from all fish ($n = 7-10$), while light symbols and lines include only those data from fish that were placed in respirometers and had $\dot{M}_{O_2}$ measured ($n = 4$). Values are expressed as means $\pm$ SE. Asterisks refer to dark symbols and denote statistically significant differences ($P < 0.05$) from the initial value at 13°C. Statistics were not performed on data in light symbols due to insufficient statistical power, although it is evident that data from these fish followed similar relationships to those determined for all fish combined.

Instruments 3320 freezing point osmometer), chloride (Haake Buchler digital chloridometer), and sodium and potassium (Cole-Parmer, model 410 single channel flame photometer) (see Ref. 21 for further details).

Statistical analyses. All fish generally remained calm during the majority of the experiments, and so it was possible to exclude active periods from the analysis such that only data from resting fish are presented. The effect of increasing temperature on mean ($\pm$ SE) hematological and cardiovascular variables was assessed with one-way repeated-measures ANOVA followed by Dunnett's post hoc test to identify individual points that were significantly different from the initial value at 13°C. Significance was considered at $P < 0.05$.

RESULTS

Morphometrics. Body mass (M_b) was related to fork length (FL) as $M_b = 0.15 \times FL - 6.65$ ($r^2 = 0.94$), where M_b is in kilograms and FL is in centimeters. Total wet heart mass (atrium, ventricle, and bulbus) was 9.29 ± 0.86 g, and relative wet heart mass was $0.238 \pm 0.004\%$ of body mass. The dry relative ventricular mass was $0.028 \pm 0.001\%$ of wet body mass, and the ventricle was composed of $53.2 \pm 0.8\%$ compact myocardium based on the ratio of the dry masses of the spongy and compact myocardial layers.

Cardiovascular, hematological, and respiratory responses to temperature change. At 13°C, routine $\dot{Q}$ was 28.5 ± 2.5 ml·min⁻¹·kg⁻¹, f_H was 58.3 ± 2.5 beats/min, and V_S was 0.49 ± 0.04 ml·beat⁻¹·kg⁻¹. The dorsal aortic and central venous blood pressures were 5.20 ± 0.22 kPa and 0.01 ± 0.03 kPa, respectively (Fig. 1). Cardiovascular values measured for the fish in the respirometers were not different from these values (Fig. 1), and routine $\dot{M}_{O_2}$ was 3.4 ± 1.0 mg·min⁻¹·kg⁻¹. Hematological variables are summarized in Fig. 2 and Table 1.

Increasing water temperature gradually and significantly increased $\dot{Q}$. This response was mediated entirely through tachycardia because V_S was statistically unchanged (Fig. 1). At 25°C, $\dot{Q}$ had increased by 97% to 56.3 ± 5.4 ml·min⁻¹·kg⁻¹ and f_H had increased by 81% to 105.3 ± 2.3 beats/min. This resulted in a Q_{10} value of 1.8 for $\dot{Q}$ and 1.6 for f_H between 13 and 25°C. Associated with the increase in $\dot{Q}$ was a significant reduction in systemic vascular resistance (R_{SYS}), which resulted in P_{DA} being unchanged during heat stress. Similarly, P_{CV} was unchanged at 17 and 21°C but increased significantly to 0.19 ± 0.05 kPa at 25°C. Cardiovascular responses of the fish in the respirometers followed the same trends. $\dot{M}_{O_2}$ increased by 152% to 8.7 ± 1.1 mg·min⁻¹·kg⁻¹ at 25°C, which represented a Q_{10} value of 2.2. The blood convection requirement ($\dot{Q}/\dot{M}_{O_2}$) decreased from 10.2 ml/mg O₂ at 13°C to 8.0 ml/mg O₂ at 25°C.

Increasing temperature to 25°C significantly decreased MCHC in arterial and venous blood from about 316 g/l to about 279 g/l, resulting from a slight increase in Hct, likely due to erythrocyte swelling, while [Hb] was maintained (Fig. 2). P_{aO_2} and P_{vO_2} tended to decrease by similar amounts, although only P_{vO_2} at 25°C was significantly lower than the initial 13°C value (Fig. 2). There was a decrease in Cv_{O_2} that was greater than the decrease in Ca_{O_2} , such that $Ca_{O_2}-Cv_{O_2}$ increased with temperature and, along with increased $\dot{Q}$, contributed to the more than doubling of $\dot{M}_{O_2}$. Interestingly, oxygen partial pressures and contents of arterial and venous blood typically scaled negatively with body mass across temperatures (Fig. 3). Plasma lactate concentration increased significantly at 25°C, while

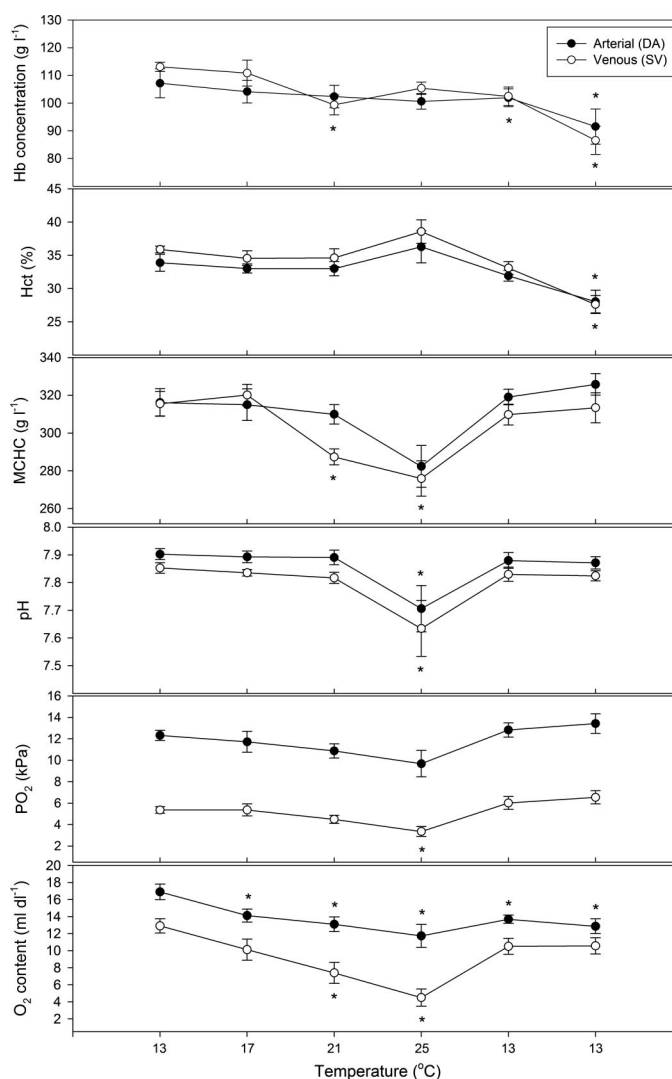


Fig. 2. Effects of acute temperature changes on hematological variables ($n = 7-10$) in Chinook salmon (*Oncorhynchus tshawytscha*). Both arterial (solid circles) and venous (open circles) blood was obtained from the dorsal aorta and the sinus venosus, respectively. The variables are hemoglobin concentration (Hb), hematocrit (Hct), mean corpuscular hemoglobin concentration (MCHC), pH, oxygen partial pressure (P_{O_2}), and oxygen content. Values are expressed as means $\pm$ SE. Asterisks denote statistically significant differences ($P < 0.05$) from the initial value at 13°C (arterial at top of figure, venous at bottom).

whole blood pH decreased significantly and there was a decrease in plasma glucose levels (Fig. 2; Table 1). Thus, despite the large increase in $\dot{M}_{O_2}$, there was evidence of insufficient oxygen delivery to the tissues at 25°C. Indeed, prominent cardiac arrhythmias at 25°C occurred in the first-, second-, and fourth-largest fish (>4 kg; Fig. 3), and one of these fish (second largest) lost equilibrium and failed to recover despite an immediate decrease in water temperature.

The progressive changes in oxygen content and partial pressure at each temperature are illustrated in the form of in vivo oxygen equilibrium curves in Fig. 4. Increasing temperature caused a modest right-shifting of the curve, and the calculated P_{50} values are given in Table 1. The estimated P_{50} values for 13°C may be underestimated because there were no data below 50% hemoglobin saturation at this temperature and

Table 1. Hematological variables of Chinook salmon (*Oncorhynchus tshawytscha*) during acute changes in water temperature

Variable	Temperature (°C)											
	13		17		21		25		13		13	
	DA	SV	DA	SV	DA	SV	DA	SV	DA	SV	DA	SV
Cortisol	229±51	218±56	178±55	175±60	205±45	200±52	220±29	236±42	250±55	231±61	297±54	240±82
Lactate	1.4±0.2	1.5±0.2	1.6±0.3	1.5±0.2	2.3±0.4	2.5±0.4	8.2±2.4*	6.8±2.3*	3.1±0.9	2.7±0.3	3.4±1.1	2.8±1.1
Glucose	4.2±0.3	4.2±0.4	3.7±0.4*	3.5±0.4*	3.1±0.3*	3.2±0.4*	3.2±0.4*	3.4±0.5*	4.0±0.3	4.0±0.2	4.0±0.3	3.9±0.3
Osmolality	289.9±2.6	290.6±2.4	283.1±5.0	282.9±3.3	286.8±4.1	284.6±3.0	295.8±3.3	298.5±2.2	292.1±2.8	288.9±2.8	290.1±3.4	290.8±1.4
Chloride	127.2±2.0	126.1±2.3	124.5±2.7	123.0±1.7	125.9±1.5	122.7±2.2	126.4±2.1	125.5±1.5	129.9±2.8	125.0±2.7	127.9±2.1	129.7±1.5
Potassium	4.6±0.3	2.9±0.2	6.0±0.4	3.9±0.5	7.0±0.7	3.6±0.4	6.0±1.3	2.9±0.4	8.1±1.5	3.0±0.4	6.0±1.3	3.1±0.4
Sodium	144.8±1.8	149.6±2.6	141.5±3.0	145.3±3.4	142.0±3.2	143.9±3.2	146.9±2.2	146.9±1.3	142.3±3.1	145.7±3.5	143.2±2.4	149.4±3.2
P ₅₀	2.95		3.71		4.27		4.95		3.39		3.39	

Values are expressed as means ± SE; $n = 7-10$. DA, dorsal aorta blood (arterial); SV, sinus venosus blood (venous). Units are cortisol (ng/ml), lactate (mmol/l), glucose (mmol/l), osmolality (mOsm/kg) chloride (mmol/l), potassium (mmol/l), sodium (mmol/l). *Significantly different from the initial value at 13°C. P₅₀ (kPa) predicted from regressions in Fig. 4, and no statistics were performed.

the P₅₀ values were derived from extrapolation rather than interpolation.

Cardiovascular, hematological and respiratory responses during recovery from heat stress. After just 1 h recovery at 13°C following the heat stress, $\dot{M}_{O_2}$, $\dot{Q}$, and f_H were all restored to their initial control values at 13°C (Fig. 1). Similarly, nearly all hematological variables, with the exception of CaO₂, [Hb], and Hct, were restored to their initial control values at 13°C after 1 h of recovery and remained at normal levels through to 2 h of recovery (Fig. 2; Table 1). There was a significant arterial hypotension because the reduction in R_{SYS} persisted throughout the second hour of recovery. P_{CV} also decreased significantly while fish were recovering at 13°C, and this resulted in a transient negative P_{CV} (-0.16 ± 0.03 kPa) at the first hour of recovery, which is indicative of the heart functioning with a *vis-a-fronte* filling mechanism.

DISCUSSION

The large wild salmon were observed to remain calm in the test apparatus, given that there were very few movement artifacts on the recordings from the blood flow probe. Indeed, the ranges for routine f_H and $\dot{M}_{O_2}$ at 13°C in the present study (54–64 beats/min and 1.9–3.4 mg·min⁻¹·kg⁻¹, respectively) compare favorably with those reported in previous studies with smaller Chinook salmon at similar or lower temperatures (27, 34, 44, 59). Although long postsurgical recovery times are desirable, this finding suggests that the relatively short recovery period (at least 10 h) for the routine surgical procedures used in the present study was an acceptable compromise between recovery time and the need to quickly perform the experiments before the Chinook salmon fully matured and deteriorated as their spawning date approached (about 3 wk after capture).

While routine f_H and $\dot{M}_{O_2}$ are comparable to other studies, the initial routine P_{DA} of 5.2 kPa is 13–62% higher than previous measurements of P_{DA} in juvenile Chinook (3.2–4.6 kPa) (27, 34, 44, 59). This systemic hypertension could reflect a stress (e.g., elevated sympathetic tone causing systemic vasoconstriction) that was not expressed as an elevated $\dot{M}_{O_2}$, or it may reflect the advanced sexual maturity of these adult male Chinook salmon. Indeed, male rainbow trout are known to become hypertensive during sexual maturation due to cardiac hypertrophy and hypervolemia (9). We measured dry relative ventricular mass as 0.028%, which converts to a wet relative

ventricular mass of 0.18% [if we assume 84% water content (53)]. This value is almost twice that measured in immature Chinook (27, 59), and it suggests a state of cardiac hypertrophy in our fish of a level previously observed in mature male rainbow trout (9, 28, 60).

Measurements of the response of systemic vascular resistance to temperature change in fish are scarce because very few studies have measured blood pressures and flows directly and simultaneously during acute temperature changes. Values of R_{SYS} at 13°C in the present study bracketed what has been reported previously for juvenile Chinook (27). The present study probably exposed Chinook salmon to temperatures very close to their upper thermal limit since we saw evidence of a mismatch between oxygen supply and demand in the form of cardiac arrhythmias, as well as acidosis and lactate appearance in the blood at 25°C. Increasing temperature to 25°C saw a gradual and significant decrease in R_{SYS} that was simultaneous with the increase in $\dot{Q}$ such that P_{DA} did not change. This response is consistent with systemic vasodilatation in muscle tissue and is consistent with total vascular resistance decreasing during an acute heat stress in the Antarctic fish *Pagothenia bernachii* (3). At temperatures closer to the optimum for rainbow trout (acute increase from 10°C through 13°C to 16°C), R_{SYS} did not change significantly but did tend to decrease with temperature (48). Combined with the decrease in P_{VO_2}, we can speculate that this vasodilatation reflects increased tissue perfusion of parallel vascular beds to reduce oxygen diffusion distances. There was a pronounced arterial and venous hypotension that persisted for at least 2 h after fish had been returned to 13°C following the acute heat stress. The arterial hypotension may be indicative of a metabolite-induced response lingering while energy stores are restored following anaerobic metabolism. A similar arterial hypotension is seen during recovery from the anaerobic effort involved in critical swimming speed tests with juvenile Chinook salmon (27). To what degree a temperature-related decrease in blood viscosity (25, 30, 31) might contribute to the decrease in R_{SYS} during heat stress is unclear but would not be a factor when 13°C conditions were restored. Regardless of the triggers or their mechanisms, all but the second largest fish (Fig. 3) recovered quickly and well from the acute heat stress. In fact, while the increase in P_{CV} at 25°C perhaps can be interpreted to indicate cardiac insufficiency at this temperature, the subambient pres-

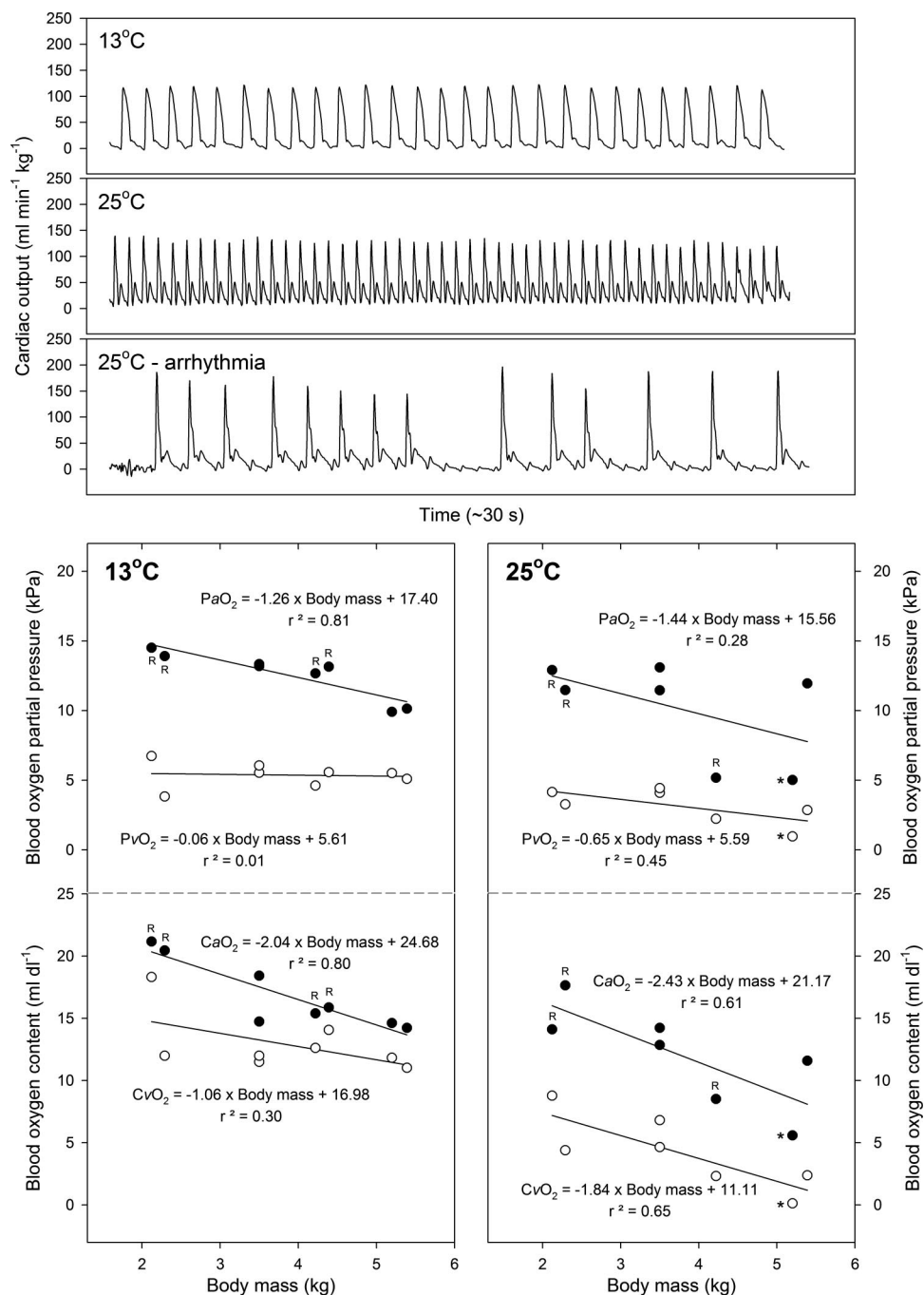


Fig. 3. *Top*: individual blood flow traces through the ventral aorta of Chinook salmon (*Oncorhynchus tshawytscha*) at 13°C, immediately upon reaching 25°C, and after being exposed to 25°C for 1 h. *Bottom*: arterial (closed circles) and venous (open circles) blood oxygen partial pressure (P_{aO_2} and P_{vO_2} , respectively) and oxygen content (CaO_2 and CvO_2 , respectively) as a function of body mass in Chinook salmon at 13°C (*left*) and when acutely exposed to 25°C (*right*) for 1 h ($n = 7-8$). R represents those fish that were housed in the respirometers (includes the labeled arterial value and the corresponding venous value at the same body mass), whereas all other fish were housed in holding tubes (see MATERIALS AND METHODS). Missing data points were due to blocked blood catheters. Asterisks indicate the individual displaying arrhythmias in the top panel of this figure.

sure on recovery is clearly indicative that the ventricular contractions are sufficiently strong to generate the *vis-a-fronte* cardiac filling that may be characteristic of the salmonid heart under routine conditions (24).

It has often been suggested that the decreasing oxygen content of the water and limited gill ventilation could contribute to the failure of gill oxygen uptake to meet the increased tissue oxygen demand with increasing temperature (5, 33, 58), although others have recently argued against this (16). The present study showed that five out of seven Chinook salmon maintained high P_{aO_2} during heat stress, even at the point where oxygen delivery to tissues was compromised (Fig. 2; Fig. 3). Consistent with our results is the finding that P_{aO_2} did

not decrease in exercising sockeye salmon at high temperatures (M. F. Steinhausen, E. Sandblom, E. J. Eliason, C. Verhille, A. P. Farrell, unpublished data). The maintenance of P_{aO_2} found for Chinook and sockeye is possible only if the conditions for oxygen diffusion across the gills and the ventilatory delivery of oxygen to the gills did not become limited under the present experimental conditions of heat stress. However, a diffusion limitation at the gills could be suggested for those larger individuals that did not maintain P_{aO_2} at 25°C (Fig. 3; Fig. 4). This may be related to increased gill diffusion distances in larger Chinook, and so the allometry of gill thickness and oxygen uptake in different species of fish would be an interesting avenue for future research.

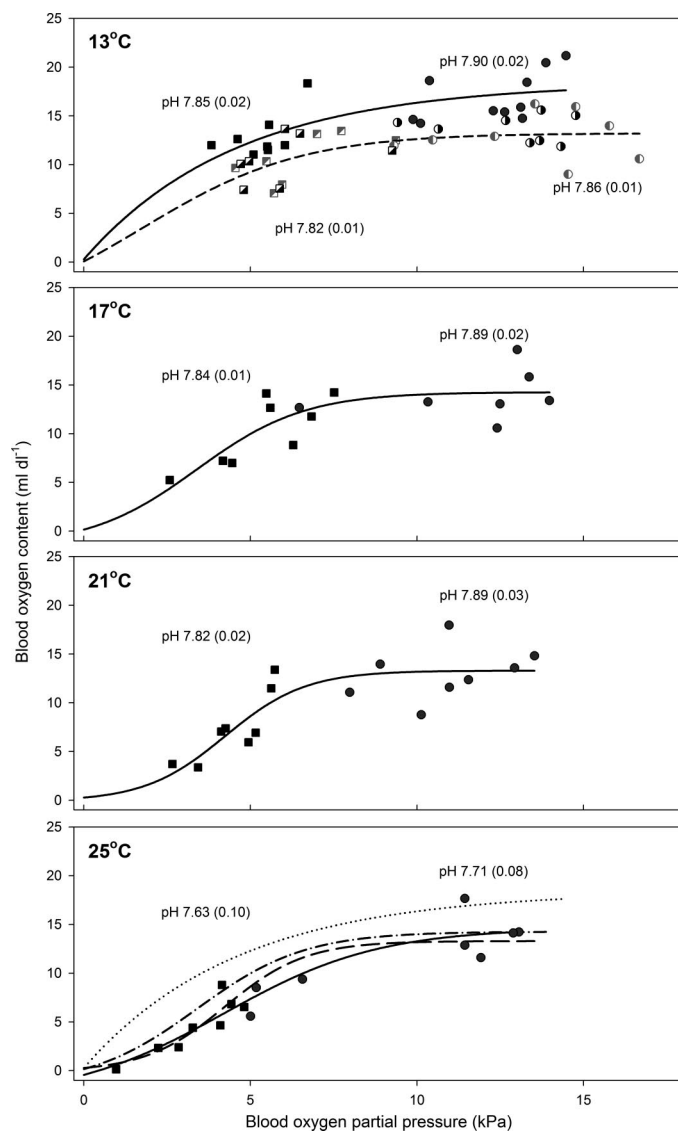


Fig. 4. Oxygen equilibrium curves at 13, 17, 21, and 25°C, as determined from arterial (circles) and venous (squares) blood samples taken from catheters at each experimental treatment. Also presented are the recovery data at 13°C; 1 h (half-filled black symbols on right) and 2 h (half-filled gray symbols on left) recovery data have been combined to construct the dashed curve. All curves were constructed using a 4-parameter sigmoid function in SigmaPlot. pH values (SE in parentheses) for arterial (right) and venous (left) data are given. All curves are presented on the 25°C panel for easy comparison: dotted line = 13°C; dashed-dotted line = 17°C; dashed line = 21°C; solid line = 25°C.

Notably, Hct, [Hb], and CaO_2 were depressed during recovery, perhaps because resequestering of red blood cells in the spleen revealed the impact of withdrawing numerous blood samples that totaled $<10\%$ of blood volume if we assume a blood volume of 4%. Evidence against this possibility lies in the fact that the apparently chronic decrease in CaO_2 occurred immediately when the fish were exposed to the first temperature increment of 17°C (Fig. 2; Fig. 4), at which point the cumulative blood volume withdrawn was small. It may be that the decrease in CaO_2 at 17°C was associated with allosteric modulation of hemoglobin-oxygen binding, but this cannot be confirmed without further experimentation. Nevertheless, *in vivo* oxygen equilibrium curves of Chinook salmon did not

substantially right-shift with increasing temperature (Fig. 4), which contrasts with the suggestion of Gollock et al. (29) and Heath and Hughes (33) that oxygen binding capacity is limited at high temperatures in Atlantic cod and rainbow trout, respectively.

Randall and Daxboeck (42) have suggested that oxygen exchange in rainbow trout gills was limited by blood perfusion during exercise. Here, we found that the increase in $\dot{Q}$ did not match the increase in $\dot{M}_{\text{O}_2}$ during heat stress in Chinook, as indicated by a reduction in the blood convection requirement, which is also consistent with a perfusion limitation at the gills during heat stress. This may be simply because $\dot{Q}$ has either reached a functional limit or a physiological limit relative to the experimental conditions. It is interesting, yet unclear why the increase in $\dot{Q}$ with temperature in this study was mediated by an increase in f_{H} , while a similar increase in $\dot{Q}$ during exercise trials at 8–10°C in juvenile Chinook was mediated primarily by an increase in V_{S} (27). Maximum $\dot{Q}$ in salmonids is reportedly able to reach $60\text{--}110\text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ (2, 19), and f_{H} to have a maximum of around 120 beats/min (20). Therefore, f_{H} may have approached a maximum during heat stress in the present study, and so it is curious why V_{S} of Chinook remained unchanged, as has been found in a number of previous studies (8, 12, 29, 57). V_{S} was not abnormally high, as maximum V_{S} is reported in the range of $1\text{ ml}\cdot\text{beat}^{-1}\cdot\text{kg}^{-1}$ for salmonids, and so this removes the possibility that V_{S} was already near its maximum. Three potential reasons come to mind why V_{S} does not increase concomitantly with f_{H} and further enhance arterial oxygen transport at elevated temperatures. First, cardiac contractility decreases with increasing contraction frequency [a negative force frequency relationship (51)] and high temperature per se (35). Therefore, end-systolic volume may increase with elevated f_{H} because of weaker ventricular contractions. Second, cardiac filling time and potentially filling pressure (i.e., P_{CV}) are negatively correlated with f_{H} and temperature (1, 47, 48). Temperature may passively increase venous compliance (32, 45, 52), possibly reducing circulating blood volume and decreasing adrenergically mediated venous vascular tone (48). However, Sandblom and Axelsson (48) demonstrated for rainbow trout that cardiac filling pressure is maintained constant due to a decrease in vascular capacitance, which mobilizes blood to the central venous compartment when f_{H} increases during a moderate (10–16°C) acute temperature increase. Also, P_{CV} was initially unchanged in Chinook salmon as f_{H} increased with temperature, suggesting that there was a compensatory decrease in venous vascular capacitance which mobilized blood to the central venous compartment and compensated for the tachycardia (47, 48). In fact, at 25°C in Chinook, P_{CV} was significantly increased, and yet V_{S} did not increase despite an increased cardiac filling pressure, an observation that is consistent with cardiac congestion and contractile insufficiency. The third possibility relates to the deleterious changes in the chemical composition of the blood that could weaken cardiac contractility *in vivo* (e.g., acidosis and hypoxemia), which has been used as an explanation why maximum $\dot{Q}$ for *in situ* working trout hearts receiving fresh perfusate is typically greater than that measured during exercise when the chemical composition of venous blood changes substantially (16, 18, 26). In a similar context, cardiac arrhythmias were present in three of the largest Chinook at 25°C, and one of these fish (second-largest) lost equilibrium and failed to

recover. This latter fish was the only individual to reach a PvO_2 as low as 1 kPa (Fig. 3), and thus it may be suggested that this is the lower PvO_2 threshold for sufficient cardiac function. This is in agreement with data from rainbow trout when exposed to hypoxia during a swimming challenge (55), and the associated limitation in cardiac function will lead to gill perfusion limitations as discussed above.

Perspectives and Significance

The present study has shown that this population of Chinook salmon relies on a combination of Q and $CaO_2 - CvO_2$ to meet the increased $\dot{M}O_2$ required during heat stress. We have uncovered some potential effects of body mass on arterial oxygen saturation, which may be linked with greater gill diffusion distances in larger fish. This is a topic that warrants further investigation over a greater body mass range, preferably with a species that grows well in excess of 5 kg. Nevertheless, it seems that tolerance to high temperature decreases with body mass, as indicated by lower arterial and venous oxygen levels. Given the switch to anaerobic metabolism and the cardiac arrhythmias observed in large fish at 25°C, we propose that the cascade of venous oxygen depletion results in a threshold value for venous oxygen partial pressure of around 1 kPa (Fig. 3). At this point, the oxygen supply to systemic and cardiac tissues is compromised such that the oxygen-deprived and acidotic myocardium becomes arrhythmic and blood perfusion through the gills and to the tissues becomes compromised. We suggest that the critical thermal maximum for resting adult Chinook salmon is mass-dependent and lies around 25°C for large fish (>4 kg) and perhaps around 27°C for smaller adult individuals. There is likely to be negligible aerobic scope at these temperatures, and so swimming fish are likely to have a reduced critical thermal maximum to maintain satisfactory aerobic scope for locomotion.

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